

Parameter Configuration Document

This document outlines the default settings, variables, and the critical **parameter mapping logic** used by the **MusicGenerator** class to translate high-level user requests into concrete model hyper-parameters.

1. Global & Default Configuration

Parameter	Value / Variable	Purpose
Model Used	facebook/musicgen-small	Chosen for stability and efficiency on the Dedicated Processing Unit (GPU).
Target Sample Rate	32000 Hz	Standard sample rate output by the MusicGen model.
Output Format	MP3 (192k bitrate)	Final required format, achieved via pydub and FFmpeg.
Tokens per Second	approx 50 tokens/second	The estimated rate used to calculate audio duration from max_new_tokens.

2. Dynamic Parameter Mapping Logic

The core intelligence is the function that converts user-friendly, semantic inputs (Energy Level, Duration) into the technical parameters required by the MusicgenForConditionalGeneration model.

User Input Parameter	Model Hyper-parameter	Mapping Formula / Range	Rationale

Energy Level $\mathbf{1-10}$	temperature (Randomness)	$0.6 + (\text{Energy} / 10) \times 0.4$	Higher energy maps to higher temperature (0.6->1.0), increasing musical variability and dynamism.
Energy Level $\mathbf{1-10}$	guidance_scale cfg_coef	$4.0 - (\text{Energy} / 10) \times 2.0$	Higher energy maps to lower CFG (4.0 -> 2.0), allowing the model more creative freedom (less strict adherence to the prompt).
Duration $\mathbf{X}$ seconds	max_new_tokens	Tokens = 50 X Duration	Controls the exact length of the generated audio sample based on a standard estimation of model token rate.

3. Audio Post-Processing Pipeline

Post-processing is executed using the pydub and FFmpeg utilities to ensure professional, polished audio output.

Post-Processing Step	Purpose	Technical Implementation

Volume Normalization	Normalizes the audio to a standard peak level.	Peak normalization targeted at approximately -3 dB (0.707 in float array), ensuring consistent playback volume.
Fade In/Out	Smoothly ramps the audio volume at the start and end.	Applied 1 second (1000 milliseconds) of smooth fading on both ends, removing abrupt starts/stops.
Format Conversion	Converts the raw audio to the final deliverable format.	Conversion of the intermediate <code>.wav</code> file to MP3 format using the configured FFmpeg encoder.